

BHOOMI OS — AGRICULTURAL EVIDENCE LAYER

Automated Sentinel-2 & Meteorological Proof Packet for PMFBY Claims

PACKET ID: PKT-1

CLAIM DATE: 27/8/2026

STATUS: VERIFIED EVIDENCE

1. FARMER & AGRISTACK REGISTRY DETAILS

Farmer Name:	Ravi Kumar	AgriStack ID:	AGR-IND-88219
Phone Number:	9876543210	Farmer ID (Local):	FRM-1
Crop Sown:	Wheat	Total Area:	2.35 Hectares

2. SENTINEL-2 L2A SPECTRAL AUDIT (COPERNICUS SATELLITE)

Canopy NDVI:	0.22 (Vegetation Index)	Baseline Expected:	0.500 NDVI
Water Index (NDWI):	0.08 (Moisture Index)	Vegetation Deficit:	56% Below Norm
Satellite Mission:	Copernicus Sentinel-2 Harmonized	Spatial Resolution:	10m Multi-spectral Pixel

3. METEOROLOGICAL TELEMETRY AUDIT (OPEN-METEO 7-DAY)

Cumulative 7d Rain:	12.5 mm	Drought Threshold:	< 20.0 mm (Deficit)
Avg Max Temp:	37.8 °C	Heat Index Stress:	HIGH (Exceeds 35°C)

4. AUTOMATED STRESS DETERMINATION & PMFBY CLAIM ADVICE

STRESS: DROUGHT

Triggered Rule: R1_drought_low_ndvi_dry_spell

Assessed Crop Loss: 45% Estimated Yield Loss Confidence Score: 88% Verified

Evidence Summary: Sensed multi-band reflectance exhibits critical vegetative moisture collapse coinciding with unbroken dry spell and high thermal radiation during reproductive crop phase.

5. CRYPTOGRAPHIC VERIFICATION & AUDIT HASH

SHA-256 EVIDENCE INTEGRITY HASH (TAMPER-EVIDENT):

0x1608d8fa3d44edfeabf3bf2d4aacc2bfeb51c25c8bf2bd8e2036089fb409561b